

HACKOUT'26

RENEWCAST

AI Renewable Energy Command Center

Forecast. Detect. Recommend. Explain.

Problem Statement	AI-Powered Renewable Generation Forecasting Platform
Theme	Renewable Energy Intelligence
Team Name	TriHack
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College	L.D. College Of Engineering

1. Problem

Solar and wind generation does not stay constant. Weather, time of day and seasonal conditions can change renewable output quickly. That makes it harder for grid operators and utilities to plan backup capacity, storage and power dispatch in advance. The problem statement therefore asks for a system that can forecast renewable output over the next 24–72 hours and flag periods of expected over- or under-generation.

Why Renewable Generation Needs Forecasting

Solar and wind generation is inherently variable and depends on changing weather, time of day and seasonal conditions. This variability makes it difficult for grid operators and utilities to plan storage, backup and power dispatch in advance.



The goal is to see the problem early enough to act, not after the mismatch has already happened.

Figure 1. The basic operational problem behind the project.

Why this matters

- Under-generation can create reliability pressure and increase the need for backup generation.
- Over-generation can lead to curtailment or wasted renewable energy.
- Late decisions are more expensive than decisions made with a few hours of warning.
- Better forecasts support financial planning for utilities, plant owners and energy traders.

Our interpretation of the problem

We are treating forecasting as only the first half of the problem. The useful part is what happens after the forecast: detect the risky periods early, explain the reason, and suggest a practical response. That is the direction we will follow in the prototype.

The official problem statement also identifies grid operators, utility companies, renewable plant owners and energy traders as the main users.

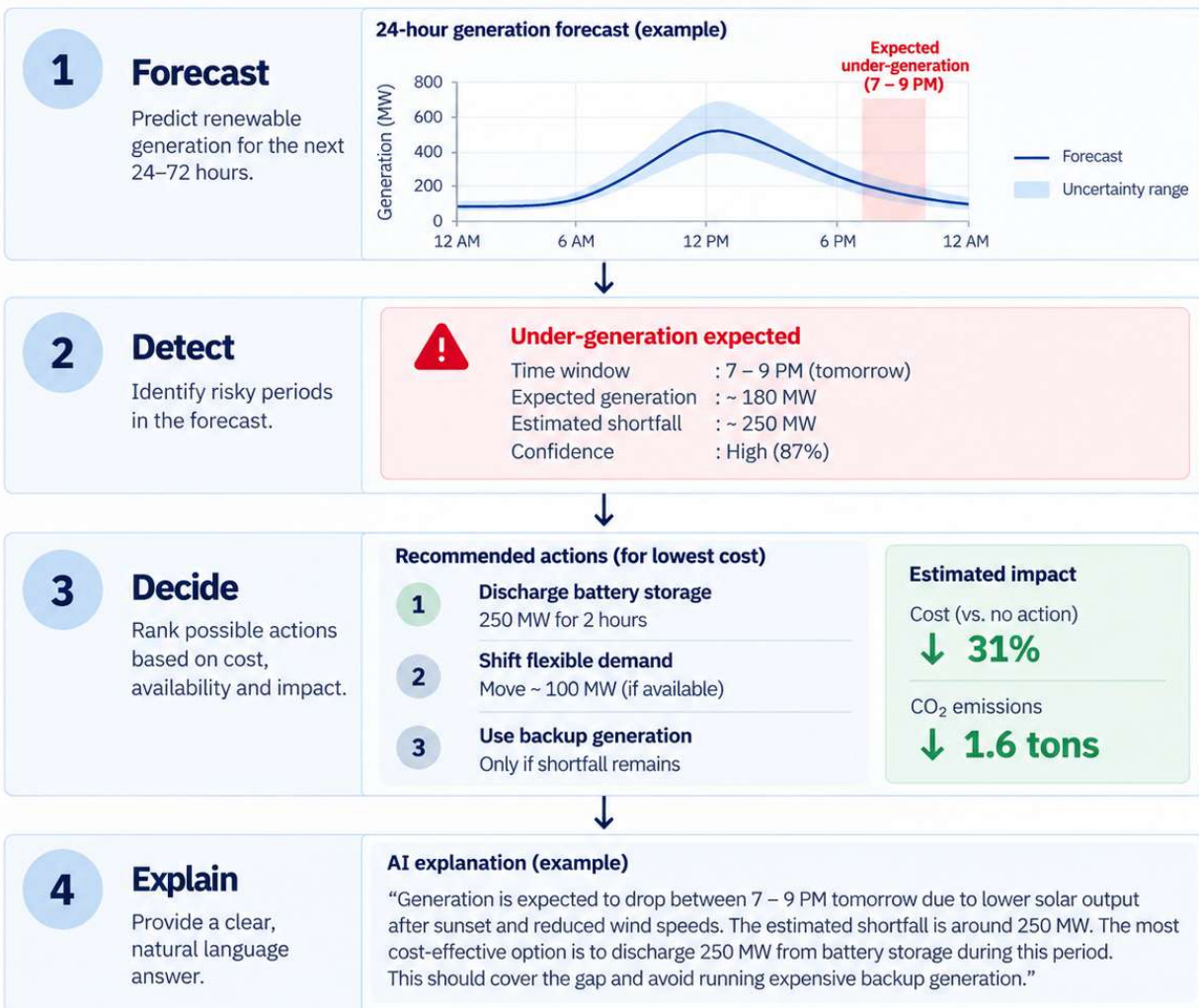
2. Users and Their Needs

User	Main need	How RenewCast helps
Grid operators	See supply risk before it becomes a grid problem.	Forecast + risk alerts + recommended response.
Utility companies	Plan backup, storage and operating schedules.	24–72 hours outlook with cost/energy impact.
Renewable plant owners	Understand future production and curtailment risk.	Site-level forecast, risk windows and action suggestions.
Energy traders	Reduce uncertainty in expected renewable supply.	Forecast trend, confidence view and major events.

Example user question

“Will I have enough renewable generation tomorrow evening, and if not, what is the least costly way to cover the gap?”

The figure below shows how RenewCast answers this question — from forecasting, to risk detection, to an actionable recommendation with an explanation.



3. Proposed Solution

RenewCast is a web-based AI decision-support platform for renewable generation forecasting. The platform combines historical generation data, site information and, where available, weather features to produce a 24–72 hour forecast. It then identifies risky intervals and suggests an operational response.

Our Proposed Solution: Forecast First, Then Decide

RenewCast combines historical generation data, site information and weather inputs to forecast renewable output, identify risky periods and recommend operational actions.

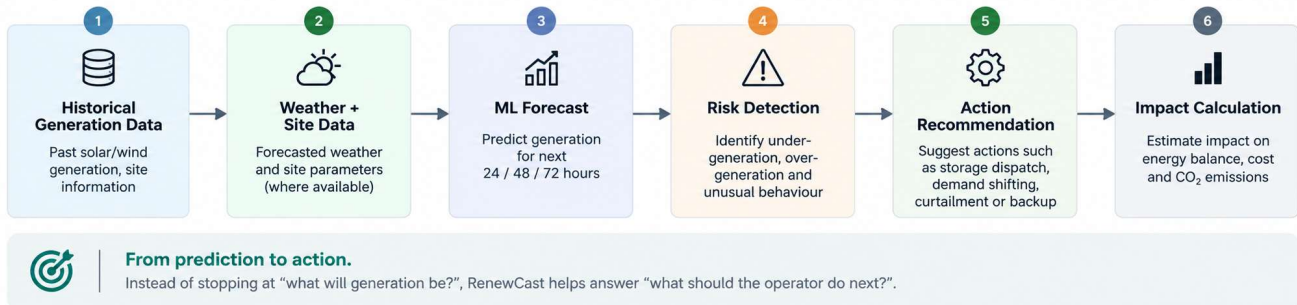


Figure 2. Solution flow.

Core idea

Stage	What happens
1. Forecast	Predict solar/wind generation for the next 24, 48 and 72 hours.
2. Detect	Flag under-generation, over-generation and unusual forecast behaviour.
3. Decide	Rank possible actions such as battery dispatch, demand shifting, curtailment or backup activation.
4. Quantify	Estimate the effect on energy, backup requirement, cost and CO ₂ .
5. Explain	Use the AI assistant to explain the result in plain language and answer operator questions.

The official problem statement explicitly asks for forecasting, over/under-generation alerts and recommended grid actions. Our added decision and explanation layers are meant to make the prototype more useful and more distinctive.

4. Problem-Solving Approach

Step 1 - Data preparation

- Read historical time-series generation data and timestamps.
- Check missing values, duplicates and unusual readings.
- Create a clean table indexed by time and renewable site.

Step 2 - Feature engineering

- Recent generation values (lag features).
- Rolling mean / rolling variation to capture short-term trends.
- Hour, day and month features to capture daily and seasonal behaviour.
- Weather variables will be integrated as model inputs where a suitable site-level source is available.

Step 3 - Forecasting model

The first model will use XGBoost or LightGBM because both are practical for a short hackathon cycle and work well with engineered tabular/time-series features. We will also keep a simple baseline so that the team can compare whether the ML model actually improves the forecast.

Step 4 - Risk and action layer

The forecast is converted into operational events. Examples are a predicted deficit during an evening period or a surplus during a high-solar window. Each event is assigned a severity and mapped to one or more possible actions.

Step 5 - Decision explanation

Gemini will receive the already-calculated forecast, risk and action data. It will explain the result, summarize the reason and answer “what-if” questions. The LLM will not be responsible for producing the core numerical forecast.

AI / ML Pipeline

From data to forecasts, and from forecasts to actionable insights.

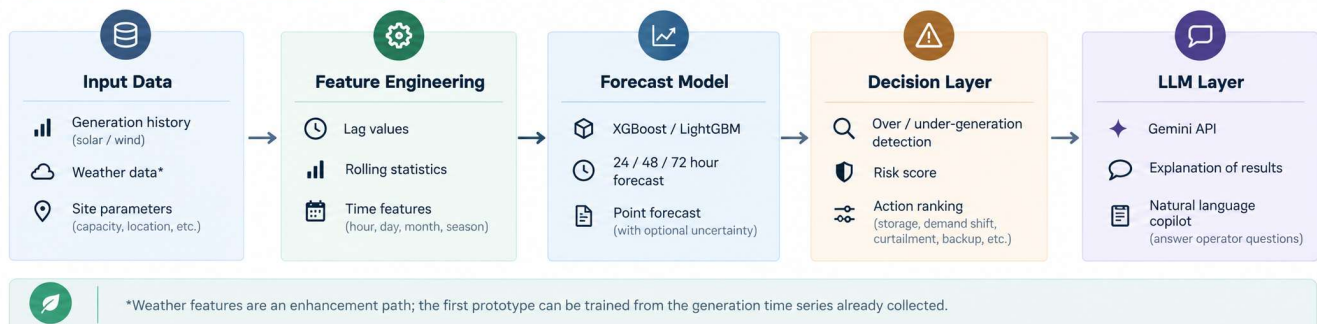


Figure 3 AI/ML pipeline.

The implementation is intentionally layered: the forecast model produces the numerical prediction, the decision engine identifies operational risk and ranks actions, and the LLM is used only to explain already-calculated results and answer operator questions. This keeps the core path measurable and testable.

Model inputs will be built from recent generation values, rolling statistics and calendar features, with weather variables added when a suitable site-level source is available. The forecasting stage will be evaluated against a simple baseline using MAE and RMSE, with additional error measures considered where appropriate.

5. Key Features

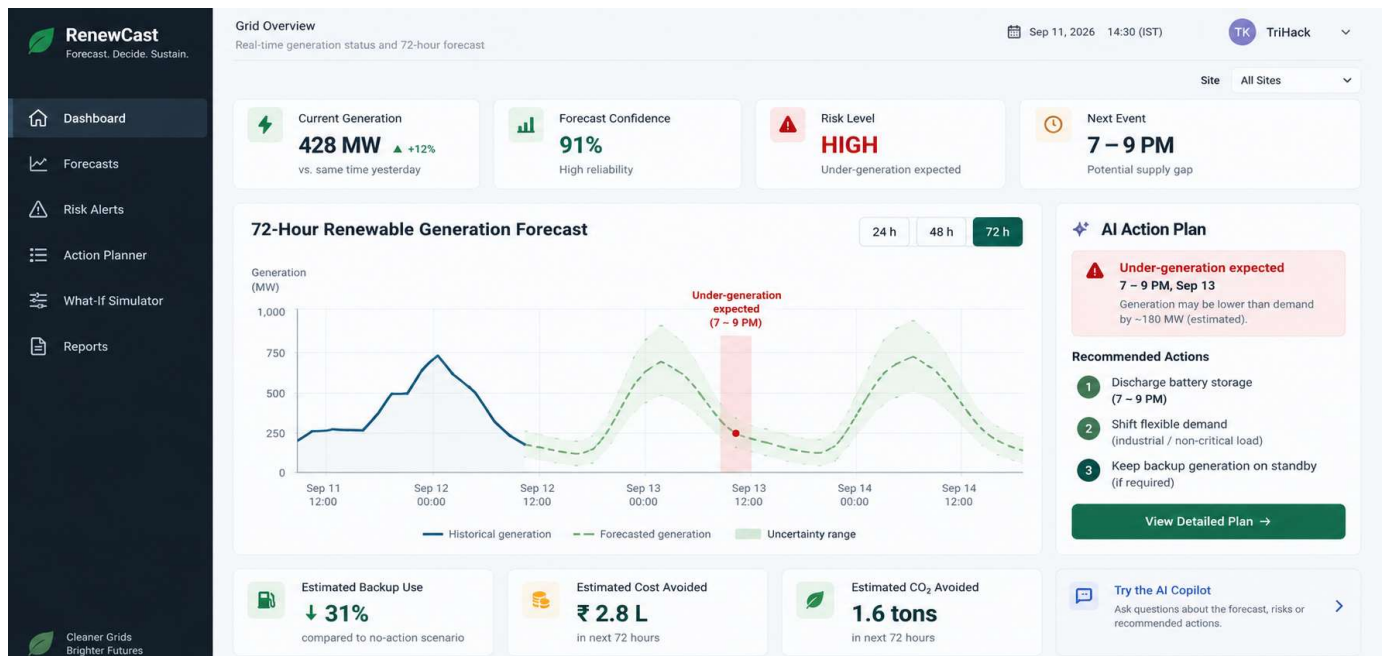


Figure 4. Early dashboard concept. Values shown are illustrative for the prototype.

24–72 hour forecasting: A clear forecast view with selectable horizons for the selected renewable site.

Over / under-generation alerts: Highlight the hours where the predicted output is likely to be above or below expected operating needs.

AI action recommendations: Recommend a practical response rather than leaving the user with only a warning.

Risk and confidence view: Show the level of concern and how confident the system is about the event.

What-if simulator: Change battery capacity, demand shifting or backup availability and see how the outcome changes.

Cost + CO₂ impact: Translate operational choices into simple numbers that a user can compare.

AI Copilot: Ask questions such as “Why is tomorrow evening risky?” and receive an explanation based on the forecast results.

Forecast history: Compare predicted and actual generation so the user can see how the model behaves over time.

6. Data and AI/ML Strategy

For the first prototype, we are using the public ARPA-E PERFORM dataset available through the U.S. Department of Energy / NREL data ecosystem. The dataset provides time-coincident solar, wind and load data at 5-minute resolution, including actual and forecast time series for several U.S. balancing areas.

Data component	Use in our project
Generation actuals	Primary training target and historical pattern.
Time index	Align the time-series and generate time-based features.
Site metadata	Keep forecasts tied to a specific renewable site.
Load data (if used)	Compare expected renewable supply with system demand.
Weather data (optional enhancement)	Add external drivers such as temperature, irradiance or wind speed.

Model plan

1. Start with a baseline forecast to establish a simple reference point.
2. Train XGBoost or LightGBM using lag, rolling and time features.
3. Evaluate with MAE, RMSE and MAPE where appropriate.
4. Compare 24-hour performance first, then extend the same pipeline to 48 and 72 hours.
5. If time allows, add weather features and compare the improvement.

Important AI separation

Our numerical layer remains deterministic and testable. The ML model predicts generation. The decision engine calculates risk and ranks actions. Gemini is used for explanation and the conversational interface. This avoids asking the LLM to invent power-generation numbers.

Dataset location note

The initial prototype uses U.S. public data because the dataset is available in a clean, high-resolution format. The platform itself is location-agnostic: the data ingestion layer can be replaced later with Indian site or grid data without changing the dashboard and decision logic.

Source: ARPA-E PERFORM datasets, National Renewable Energy Laboratory (NREL), public access.

<https://catalog.data.gov/dataset/arpa-e-perform-datasets>

7. Technical Architecture and Technology Stack

System Architecture

A modular and scalable architecture to go from data to actionable insights.

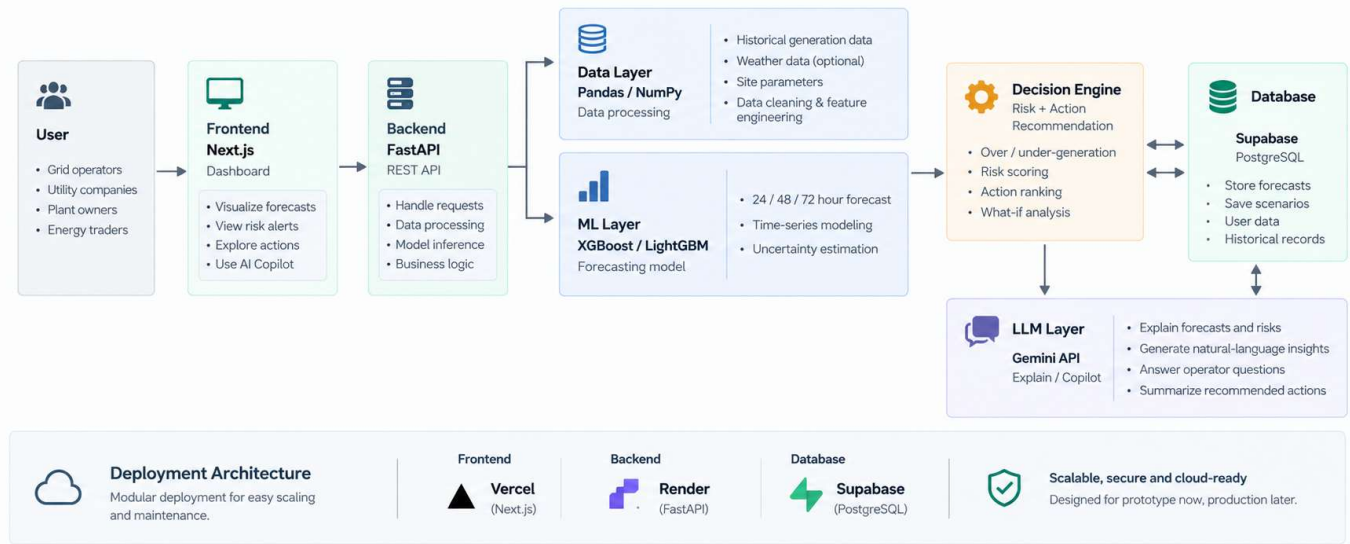


Figure 5. Planned system architecture.

Layer	Technology	Purpose
Frontend	Next.js, React, TypeScript	Dashboard, charts, forms and simulator.
UI	Tailwind CSS, shadcn/ui, Framer Motion	Clean interface and interaction.
Backend	Python, FastAPI	APIs, validation and business logic.
Data / ML	Pandas, NumPy, XGBoost / LightGBM	Preprocessing, feature engineering and forecasting.
AI assistant	Gemini API	Explanations, summaries and natural-language questions.
Database	Supabase / PostgreSQL	User, site, forecast and scenario data.
Charts	Recharts	Forecast, comparison and impact graphs.
Deployment	Vercel + Render/Railway	Simple prototype deployment.

Why this stack

The stack is intentionally simple. It keeps the ML work in Python, the interface in a modern React stack, and the LLM isolated behind a backend API. That reduces integration risk within the 30-hour build window.

8. What Makes RenewCast Different

